Statistics for Data Analysis

A new, interactive DataOut learning pat

4 tiers x 12 lessons + a gating exam per tier = 48 lessons total, built around 8 reusable interactive widgets (Recharts-based) rather than markdown-and-code alone. Positioned as the conceptual/intuition layer; it does not replace the existing Python 'Statistics and scientific computing' section, which stays focused on doing the computation in SciPy/statsmodels.

This document is for review only -- no lesson, component, or page code has been written yet.

Tier 1 - Descriptive Statistics & Exploring Data

Free tier, entry point.

1. What Statistics Actually Does

learn

Population vs. sample, categorical vs. numerical data.

2. Measures of Center: Mean, Median, Mode

widget: Outlier & Center Explorer

Drag points on a number line; mean/median/mode react live; an outlier pulls the mean but not the median.

3. Measures of Spread: Range, Variance, Standard Deviation

code-verify (numpy)

4. The Five-Number Summary & Box Plots

widget: Outlier & Center Explorer (box-plot mode)

5. Z-Scores & Standardization

code-verify

6. Visualizing Distributions: Histograms

widget: Histogram Shape Explorer

Slider reshapes a dataset; histogram + mean/median redraw live.

7. Shape of a Distribution: Skewness & Kurtosis

widget: Histogram Shape Explorer (skew mode)

8. Covariance & Correlation: A First Look

learn

Deeper dive deferred to Tier 4.

9. Summarizing Categorical Data

learn + quiz

10. Outliers: Finding and Handling Them

code-verify (IQR rule, pandas)

11. Simpson's Paradox & Misleading Averages

learn + quiz

Conceptual trap lesson.

12. Tier 1 Workshop: Explore a Real Dataset

code practice

Ties descriptive stats together on a real dataset.

Exam: Descriptive Statistics - mixed mcq + code, 70% pass.

Tier 2 - Probability & Distributions

1. What Is Probability?

learn

Sample space, events.

2. Independent vs. Dependent Events

learn + quiz

3. The Normal Distribution & the Empirical Rule

widget: Distribution Explorer (normal mode)

Mean/sigma sliders, 68-95-99.7 shading.

4. Z-Scores Revisited: Using the Normal Curve

code-verify (scipy.stats.norm)

5. The Binomial Distribution

widget: Distribution Explorer (binomial mode)

n/p sliders.

6. The Poisson Distribution

widget: Distribution Explorer (Poisson mode)

Lambda slider.

7. The Uniform Distribution & Why Shape Matters

widget: Distribution Explorer (uniform mode)

8. Simulating Randomness: The Law of Large Numbers

code practice (numpy coin-flip sim)

9. The Central Limit Theorem

widget: CLT Simulator [FLAGSHIP]

Pick a skewed population; sample-size slider; sampling distribution of the mean converges to normal.

10. Why the CLT Is the Most Important Idea in Statistics

learn

Builds on lesson 9's widget.

11. Recognizing Distributions in Real Data

learn + quiz

Pattern-matching exercise.

12. Tier 2 Workshop: Simulate and Compare Distributions

code practice

Exam: Probability & Distributions - mixed mcq + code, 70% pass.

Tier 3 - Inferential Statistics

Premium tier.

1. From Sample to Population: Why Inference Exists

learn

2. Sampling Error & the Standard Error

code-verify

3. Confidence Intervals: What "95% Confident" Really Means

widget: Confidence Interval Simulator [FLAGSHIP]

Repeated sampling draws many 95% CIs; shows what fraction actually capture the true mean.

4. Margin of Error & Sample Size

widget: Confidence Interval Simulator (sample-size mode)

5. Null and Alternative Hypotheses

learn

6. p-Values: The Most Misunderstood Number in Statistics

widget: Hypothesis Test Visualizer [FLAGSHIP]

Shade the rejection region; slide the observed statistic; p-value updates live.

7. One-Sample and Two-Sample t-Tests

code-verify (scipy.stats.ttest_ind)

8. Type I and Type II Errors

widget: Hypothesis Test Visualizer (error-region mode)

9. Statistical Power: Why Sample Size Matters

learn + quiz

10. Significance Level Misuse & p-Hacking

learn + quiz

Conceptual trap lesson.

11. Bootstrapping: Resampling Your Way to Confidence

code practice (numpy bootstrap CI)

12. Tier 3 Workshop: Design and Run an A/B Test

code practice

Capstone for the tier.

Exam: Inferential Statistics - mixed mcq + code, 70% pass.

Tier 4 - Relationships & Real-World Analysis

Premium tier.

1. Correlation vs. Causation

learn

Classic conceptual lesson.

2. The Pearson Correlation Coefficient

widget: Correlation & Regression Playground [FLAGSHIP]

Drag scatter points or slide correlation strength; r updates live.

3. Simple Linear Regression: Fitting a Line

widget: Correlation & Regression Playground (regression-line mode)

4. Residuals & R^2: How Good Is the Fit?

code-verify (numpy.polyfit, R^2)

5. A Gentle Intro to Multiple Regression

learn

6. Comparing Groups: An Intuition for ANOVA

learn + quiz

7. The Chi-Square Test for Categorical Relationships

code-verify (scipy.stats.chi2_contingency)

8. Statistical Fallacies: Survivorship Bias & Regression to the Mean

learn + quiz

Conceptual trap lesson.

9. Multiple Comparisons & Why More Tests Means More False Positives

learn

10. Choosing the Right Test

widget: Test Selector Decision Tree [FLAGSHIP, non-chart]

Click-through: data type to groups to recommended test.

11. Communicating Results Honestly

learn

Soft-skills lesson on write-ups.

12. Tier 4 Capstone: A Full Statistical Analysis on Real Data

code practice

Ties descriptive + inferential + regression together.

Exam: Relationships & Real-World Analysis - mixed mcq + code, 70% pass.

The 8 interactive widgets (built once, reused across lessons)

Outlier & Center Explorer	Tier 1 (x2)
Histogram Shape Explorer	Tier 1 (x2)
Distribution Explorer	Tier 2 (x4) - normal/binomial/Poisson/uniform modes
CLT Simulator	Tier 2 (x1)
Confidence Interval Simulator	Tier 3 (x2)
Hypothesis Test Visualizer	Tier 3 (x2)
Correlation & Regression Playground	Tier 4 (x2)
Test Selector Decision Tree	Tier 4 (x1) - non-chart, click-through

Gating (seed defaults -- trivially adjustable later via premiumConfig.js)

Tier 1 (Descriptive Statistics) -- free, entry point

Tier 2 (Probability & Distributions) -- free

Tier 3 (Inferential Statistics) -- premium

Tier 4 (Relationships & Real-World Analysis) -- premium